Data Collection and Preprocessing Phase

Date	4 April 2026
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Project Title	Restaurant Recommendation System
Maximum Marks	2 Marks

Data Quality Report Template

Data Source	Data Quality Issue	Severity	Resolution Plan

Dataset (Restaurant reviews and metadata)	Missing values in fields like restaurant name, location, or ratings	Moderate	Perform data imputation using techniques like mean/mode for numeric values and most frequent value for categorical data. Alternatively, remove rows with critical missing fields.
Dataset (User reviews) Dataset (Timestamp field)	Duplicate user review entries Inconsistent or incorrect timestamp format	Low Moderate	Remove duplicate records using drop_duplicates() in pandas or SQL DISTINCT queries. Use datetime parsing libraries (e.g., pandas.to_datetime) to standardize all date/time fields.

Dataset (Restaurant metadata)	Inconsistent formats (e.g., location written in different ways like "NY", "New York")	Moderate	Apply data standardization techniques, using string functions or regex patterns to unify the format.

Dataset (User preferences)	Sparse data or insufficient user history	High	Implement fallback strategies such as popularity-based or content-based recommendations when user data is lacking.